# RS-3D: Experiments 0 and 1, full results

What we tested, every number we measured, and what comes next · for Dawar, Shubh and Sujal · 29 October 2026

### The short version

- **The problem is real (EXP0).** 3D AI models make bigger mistakes on rolling-shutter photos. In fast motion, π³ is **2.6× worse**, VGGT 1.5× and MapAnything 1.2×.
- **The problem can be fixed (EXP1).** Straightening the bent photos with the known camera motion removes **82–99%** of the extra error in fast motion (turning only), and **90% or more** when walking is corrected too. Result: **GO** for all three models.
- **The catch.** EXP1 used the rig's gyroscope and the true camera path. A real system often has neither, and fixing photos with a gyroscope is very likely already published. Our paper must fix the bending **from the images alone**.
- **Next: EXP2.** Can a model work out the motion from its own first attempt and fix itself? The answer decides which paper we write.

### The two experiments at a glance

| | EXP0: gap test | EXP1: fix with known motion |
|---|---|---|
| Question | Are models worse on rolling-shutter photos, and worse when moving faster? | If we straighten the photos using the true motion, does the extra error go away? |
| Answer | Yes for VGGT and π³ (1.5–2.6× in fast motion). MapAnything barely (1.16×). | Yes. 82–99% removed with turning only, 90% or more with turning and walking. |
| Pre-agreed verdict | **GRAY**, because slow clips were also worse. Explained by EXP1. | **GO** for all three models |
| Run time | about 1 h 50 min | about 2 h 02 min |

All numbers in this report were **measured**. The few planning estimates are marked *projected*. Tables are generated directly from the result files in `results/exp0` and `results/exp1`.

### Contents


1. The problem, in one picture
2. What we used: data, models, scores
3. EXP0: is there a problem?
4. EXP1: can it be fixed if we know the motion?
5. Where we stand
6. The road ahead, and why
7. Appendices: glossary, run facts, changes from the plan, extra statistics, technical charts, files

# 1. The problem, in one picture

Most cameras in phones, drones and robots do not take the whole photo at once. They read it **row by row, top to bottom**, over about 30 milliseconds. This is a **rolling shutter**. If the camera moves during those 30 ms, each row is taken from a slightly different position, so straight things come out slanted or wobbly.

3D AI models such as VGGT, π³ and MapAnything assume every photo was taken in a single instant. With bent photos they misjudge where the camera was and how the scene is shaped.

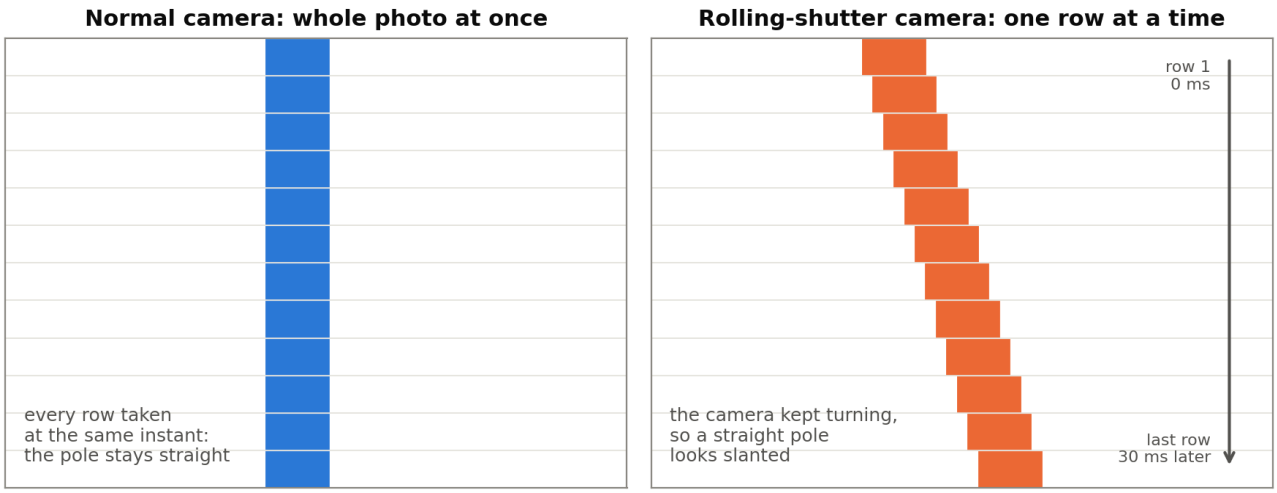



*Figure 1. A straight pole photographed while the camera turns. Left: all rows at once. Right: one row at a time.*

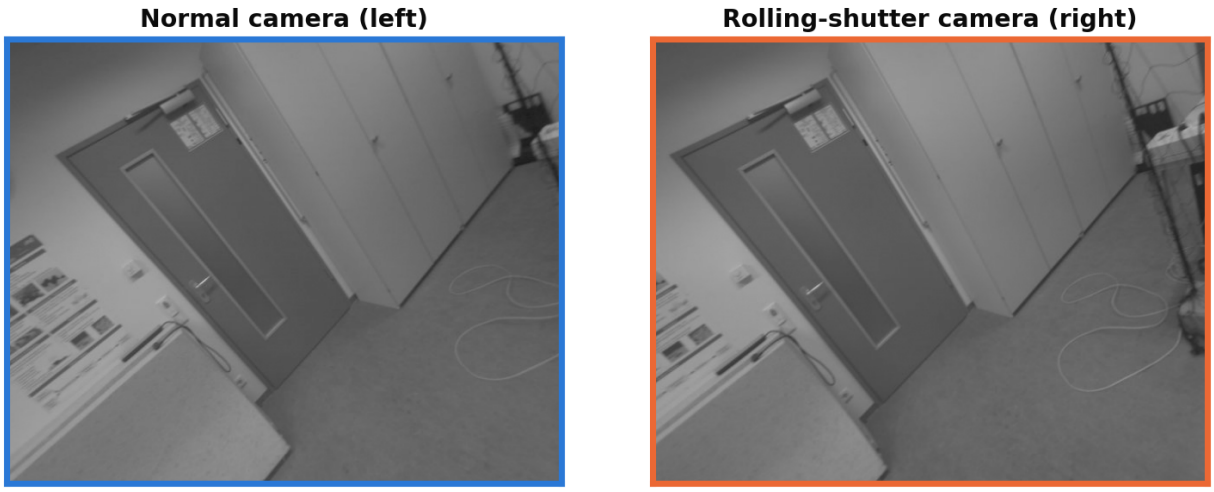



*Figure 2. Real photos from our dataset, taken at the same instant while the camera turned at about 200° per second. Part of the difference is because the two cameras sit 11 cm apart.*

> **Our project idea:** teach the 3D model that each photo is captured over time. The model also predicts how the camera was moving during the photo, and uses that to un-bend the geometry inside the model. When the camera is still, nothing changes, so normal photos are unaffected by design.

# 2. What we used

## 2.1 The dataset: TUM Rolling-Shutter (Schubert et al., IROS 2019)

| Part | What it is |
|---|---|
| Two cameras, side by side | Triggered at exactly the same moment, with the same exposure (2 ms). **Camera 1 = normal** (whole photo at once). **Camera 2 = rolling shutter** (30.18 ms from top row to bottom row). |
| Recordings | 10 indoor recordings, about 8 minutes in total, 20 photos per second, 1280×1024 pixels. |
| Gyroscope | Motion sensor: how fast the rig turns, 200 times per second. |
| Motion capture | Ceiling cameras track the rig 120 times per second. This gives the **true path**, our answer key. |
| Fairness recording | One extra recording in which **both** cameras took normal photos. It tells us how different the two cameras are, regardless of rolling shutter. |

## 2.2 The three AI models (used as-is: photos only, no hints, no training)

| Model | Notes |
|---|---|
| MapAnything (Meta) | Apache-licence weights. Originally planned as our main model. |
| VGGT-1B (Meta and Oxford) | Widely used feed-forward 3D model. |
| π³ (“Pi3”) | Newer model (2025). The most accurate of the three on normal photos. |

Each model receives 16 photos and returns where the camera was for each photo, plus the 3D shape of the scene.

## 2.3 Clips and speed groups

| Item | Value |
|---|---|
| Clip | 16 photos, every 3rd photo (2.25 s of walking). A new clip starts every 8 photos. |
| Number of clips | 1,139 made, **1,124 kept**. 15 were dropped because the rig barely moved (under 20 cm) or the answer key had gaps. The same clips are used in every experiment. |
| Speed groups (by gyroscope) | **slow**: under 29°/s (0.5 rad/s), 56 clips. **medium**: 29–86°/s, 648 clips. **fast**: 86°/s or more (1.5 rad/s), 420 clips. |
| Turning speed range | 6–206°/s across clips (median 68°/s). |

## 2.4 How we score

| Score | Meaning |
|---|---|
| **Path error** (main score, “ATE”) | How far the model's camera positions are from the true ones, on average, in cm. The guess is first lined up and scaled to the true path, because the models cannot know real-world size. |
| Turning error (“RPE-rot”) | How wrong the model is about how much the camera turned between neighbouring photos, in degrees. |
| Step error (“RPE-trans”) | How wrong the model is about how far the camera moved between neighbouring photos, in cm. |
| Times worse | Rolling-shutter error ÷ normal error, using the middle value (median) of each group. |
| Share of extra error removed | (bent − straightened) ÷ (bent − normal). 0% = no help. 100% = back to the normal camera's level. |
| 95% range | We resampled the clips 1,000 times. The true value lies in this range with 95% confidence. |
| p | Chance of seeing a difference this consistent by pure luck. “< 0.001” means almost certainly real. |

# 3. EXP0: is there a problem?

> Do 3D AI models make bigger mistakes on rolling-shutter photos than on normal photos of the same moment, and do the mistakes grow when the camera moves faster?

## 3.1 What we did

1. Prepared both cameras' photos identically: the same lens correction to one shared lens model and the same brightness curve. That way, only the shutter type differs.
2. Ran each model on every clip twice: once with camera 1 (normal) and once with camera 2 (rolling shutter). That is 3 models × 1,124 clips × 2 cameras = 6,744 runs.
3. Scored every run against the true path, then compared the cameras within each speed group.
4. Ran the same pipeline on the fairness recording (both cameras normal): 112 clips, split into thirds by speed.

## 3.2 Main result: how much worse on the rolling-shutter camera

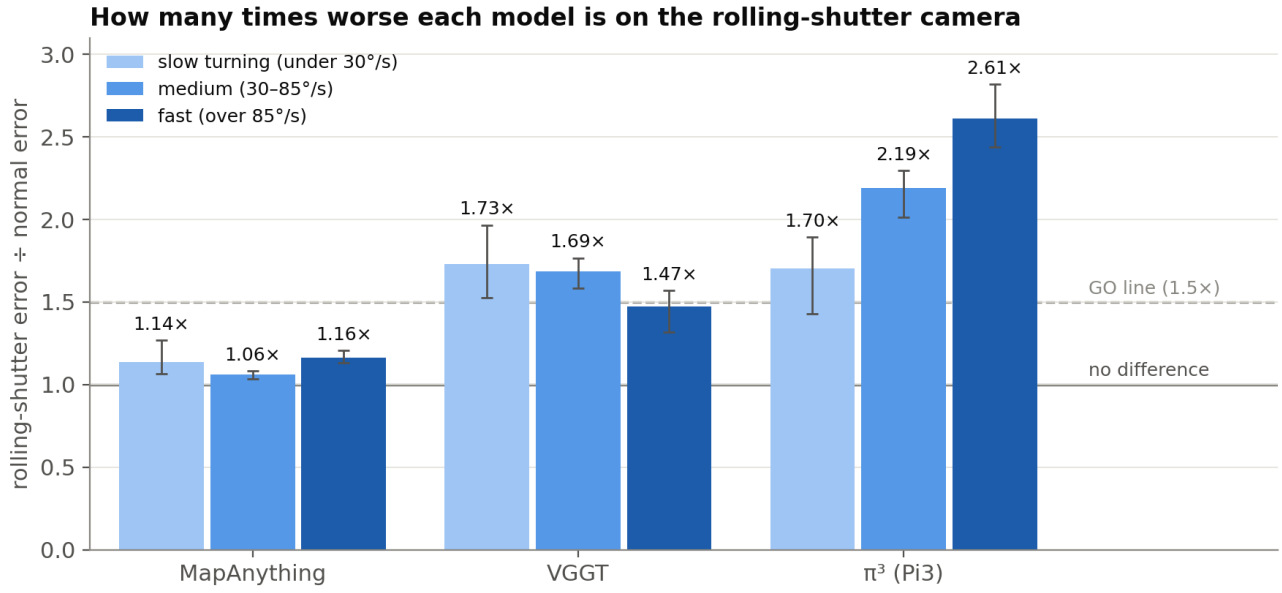



*Figure 3. Times worse (path error) on the rolling-shutter camera, by speed group. Thin lines show the 95% range. Dashed line: the 1.5× GO line.*

## Table 3.1. Path error (cm) by model and speed

| Model | Speed | Clips | Normal camera | Rolling shutter | Times worse [95% range] | Clips worse on RS | p |
|---|---|---|---|---|---|---|---|
| MapAnything | slow | 56 | 3.42 cm | 3.89 cm | **1.14×** [1.06, 1.27] | 66% | < 0.001 |
| | medium | 648 | 6.60 cm | 7.01 cm | **1.06×** [1.03, 1.08] | 61% | < 0.001 |
| | fast | 420 | 9.33 cm | 10.85 cm | **1.16×** [1.13, 1.21] | 80% | < 0.001 |
| VGGT | slow | 56 | 1.36 cm | 2.35 cm | **1.73×** [1.53, 1.97] | 98% | < 0.001 |
| | medium | 648 | 2.29 cm | 3.86 cm | **1.69×** [1.58, 1.77] | 99% | < 0.001 |
| | fast | 420 | 6.60 cm | 9.72 cm | **1.47×** [1.32, 1.57] | 88% | < 0.001 |
| π³ | slow | 56 | 0.77 cm | 1.32 cm | **1.70×** [1.43, 1.89] | 95% | < 0.001 |
| | medium | 648 | 1.03 cm | 2.26 cm | **2.19×** [2.01, 2.30] | 98% | < 0.001 |
| | fast | 420 | 1.95 cm | 5.10 cm | **2.61×** [2.44, 2.82] | 83% | < 0.001 |

"Clips worse on RS" = share of clips where the rolling-shutter run had the larger error.

- **π³** gets steadily worse as speed rises: 1.70× → 2.19× → **2.61×**.
- **VGGT** is about 1.5–1.7× worse at every speed.
- **MapAnything** is only 1.06–1.16× worse. It is also 2–6× less accurate than the others on normal photos, which can hide the effect.

## 3.3 Error against speed, in centimetres

Error grows with speed, and the rolling-shutter camera (orange) is always worse. Each panel has its own scale.

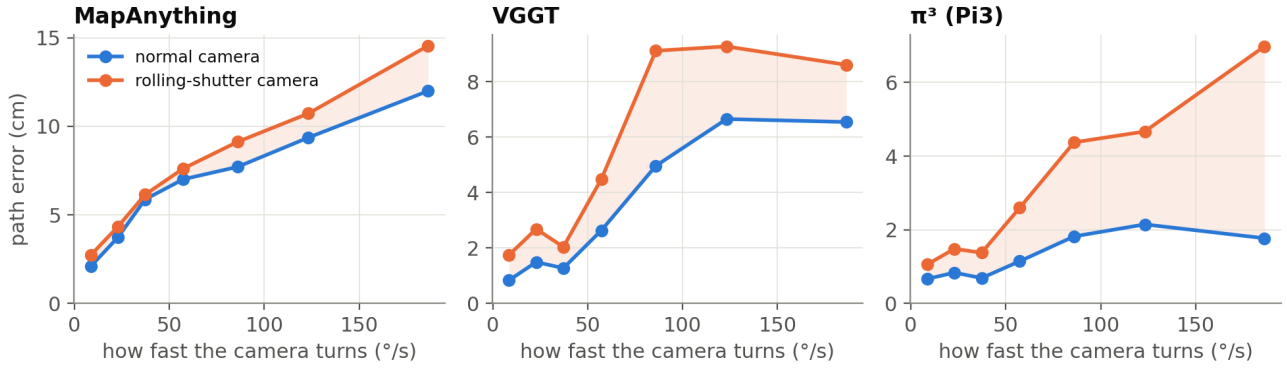



*Figure 4. Typical path error against turning speed. Blue = normal camera, orange = rolling shutter. The shaded area is the extra error.*

## Table 3.2. Finer speed steps (path error)

| Model | Turning speed | Clips | Normal camera | Rolling shutter | Extra error | Times worse |
|---|---|---|---|---|---|---|
| MapAnything | 0–17°/s | 19 | 2.10 cm | 2.73 cm | +0.64 cm | 1.30× |
| | 17–29°/s | 37 | 3.73 cm | 4.34 cm | +0.60 cm | 1.16× |
| | 29–46°/s | 308 | 5.88 cm | 6.15 cm | +0.27 cm | 1.05× |
| | 46–69°/s | 205 | 7.01 cm | 7.62 cm | +0.60 cm | 1.09× |
| | 69–103°/s | 263 | 7.71 cm | 9.13 cm | +1.41 cm | 1.18× |
| | 103–143°/s | 180 | 9.36 cm | 10.72 cm | +1.36 cm | 1.15× |
| | 143–229°/s | 112 | 11.99 cm | 14.55 cm | +2.56 cm | 1.21× |
| VGGT | 0–17°/s | 19 | 0.84 cm | 1.75 cm | +0.91 cm | 2.09× |
| | 17–29°/s | 37 | 1.49 cm | 2.69 cm | +1.19 cm | 1.80× |
| | 29–46°/s | 308 | 1.27 cm | 2.04 cm | +0.77 cm | 1.60× |
| | 46–69°/s | 205 | 2.64 cm | 4.50 cm | +1.86 cm | 1.70× |
| | 69–103°/s | 263 | 4.97 cm | 9.12 cm | +4.15 cm | 1.84× |
| | 103–143°/s | 180 | 6.66 cm | 9.28 cm | +2.62 cm | 1.39× |
| | 143–229°/s | 112 | 6.55 cm | 8.61 cm | +2.06 cm | 1.32× |
| π³ | 0–17°/s | 19 | 0.67 cm | 1.06 cm | +0.39 cm | 1.59× |
| | 17–29°/s | 37 | 0.84 cm | 1.48 cm | +0.65 cm | 1.77× |
| | 29–46°/s | 308 | 0.69 cm | 1.38 cm | +0.69 cm | 1.99× |
| | 46–69°/s | 205 | 1.15 cm | 2.60 cm | +1.45 cm | 2.27× |
| | 69–103°/s | 263 | 1.82 cm | 4.38 cm | +2.55 cm | 2.40× |
| | 103–143°/s | 180 | 2.15 cm | 4.67 cm | +2.52 cm | 2.17× |
| | 143–229°/s | 112 | 1.77 cm | 6.98 cm | +5.21 cm | 3.94× |

For π³, the extra error grows from +0.4 cm at the slowest speeds to +5.2 cm at the fastest (3.9× worse).

## 3.4 The other two scores (Tables 3.3 and 3.4)

| Model | Speed | Clips | Normal camera | Rolling shutter | Times worse [95% range] | Clips worse on RS | p |
|---|---|---|---|---|---|---|---|
| MapAnything | slow | 56 | 0.37° | 0.37° | **1.01×** [0.94, 1.13] | 57% | 0.17 |
| | medium | 648 | 0.56° | 0.61° | **1.10×** [1.06, 1.13] | 79% | < 0.001 |
| | fast | 420 | 1.22° | 1.54° | **1.26×** [1.21, 1.31] | 85% | < 0.001 |
| VGGT | slow | 56 | 0.22° | 0.32° | **1.47×** [1.31, 1.52] | 100% | < 0.001 |
| | medium | 648 | 0.30° | 0.46° | **1.54×** [1.50, 1.60] | 99% | < 0.001 |
| | fast | 420 | 0.67° | 1.14° | **1.68×** [1.58, 1.79] | 97% | < 0.001 |
| π³ | slow | 56 | 0.18° | 0.24° | **1.30×** [1.16, 1.43] | 89% | < 0.001 |
| | medium | 648 | 0.22° | 0.36° | **1.61×** [1.57, 1.67] | 98% | < 0.001 |
| | fast | 420 | 0.42° | 0.76° | **1.80×** [1.72, 1.90] | 95% | < 0.001 |

*Table 3.3. Turning error (degrees between neighbouring photos). For this score, all three models get steadily worse as speed rises.*

| Model | Speed | Clips | Normal camera | Rolling shutter | Times worse [95% range] | Clips worse on RS | p |
|---|---|---|---|---|---|---|---|
| MapAnything | slow | 56 | 2.40 cm | 3.28 cm | **1.37×** [1.21, 1.49] | 86% | < 0.001 |
| | medium | 648 | 4.99 cm | 5.51 cm | **1.10×** [1.07, 1.14] | 79% | < 0.001 |
| | fast | 420 | 9.09 cm | 10.62 cm | **1.17×** [1.14, 1.20] | 76% | < 0.001 |
| VGGT | slow | 56 | 0.91 cm | 1.80 cm | **1.98×** [1.83, 2.17] | 100% | < 0.001 |
| | medium | 648 | 1.48 cm | 2.73 cm | **1.84×** [1.73, 1.92] | 100% | < 0.001 |
| | fast | 420 | 4.64 cm | 7.33 cm | **1.58×** [1.51, 1.66] | 96% | < 0.001 |
| π³ | slow | 56 | 0.69 cm | 1.22 cm | **1.78×** [1.58, 1.97] | 100% | < 0.001 |
| | medium | 648 | 0.84 cm | 1.84 cm | **2.19×** [2.10, 2.29] | 100% | < 0.001 |
| | fast | 420 | 1.79 cm | 3.84 cm | **2.15×** [2.01, 2.26] | 91% | < 0.001 |

*Table 3.4. Step error (cm between neighbouring photos).*

## 3.5 One example: the fastest clip

The fastest 2-second clip (about 200°/s), flattened to a 2D view. Same moment, same model.

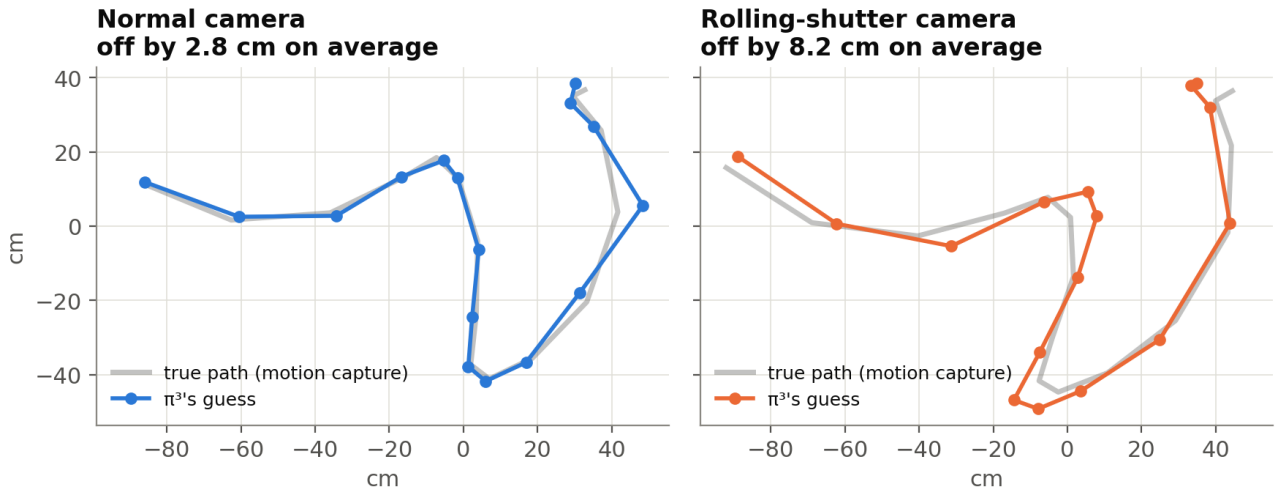



*Figure 5. Grey = true path. Coloured dots = where π³ thinks the camera was for each of the 16 photos. Normal photos: 2.8 cm off. Rolling-shutter photos: 8.2 cm off.*

### 3.6 Fairness check: is camera 2 just a worse camera?

Camera 2 has its own lens, sensor and position. In the fairness recording, **both cameras take normal photos**. If camera 2 were simply worse, it would lose there too.

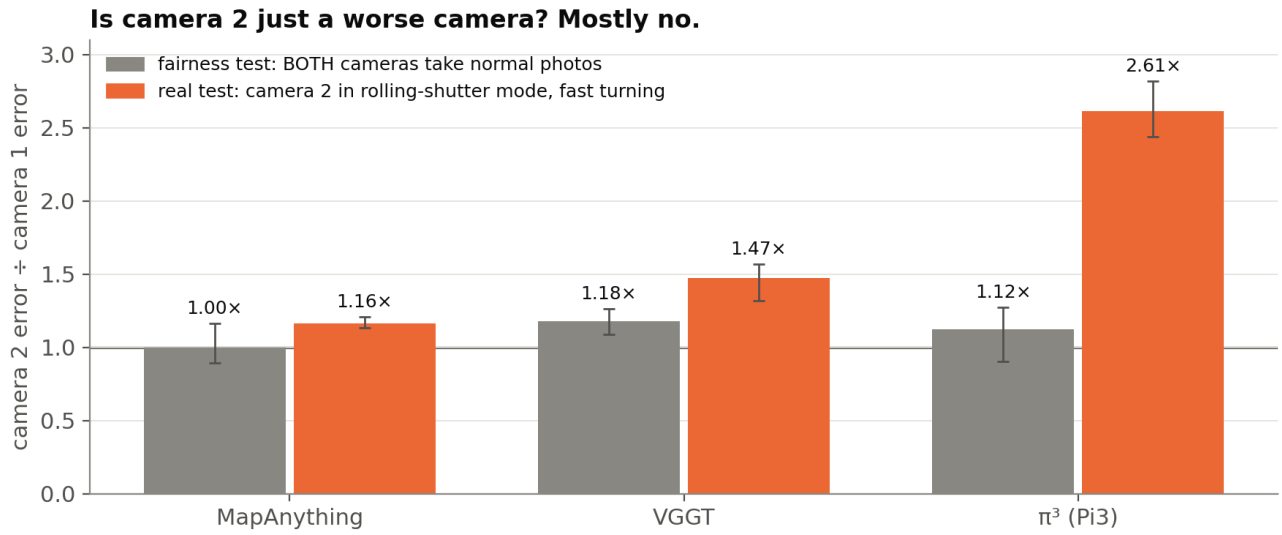



*Figure 6. Grey: both cameras normal (all clips). Orange: the real test, fast turning.*

### Table 3.5. Fairness recording, path error (both cameras in normal mode)

| Model | Speed third | Clips | Camera 1 | Camera 2 | Camera 2 ÷ camera 1 [95% range] | p |
|---|---|---|---|---|---|---|
| MapAnything | slow | 37 | 2.45 cm | 2.61 cm | **1.07×** [0.88, 1.34] | 0.12 |
| | medium | 37 | 1.71 cm | 1.92 cm | **1.12×** [0.98, 1.34] | 0.57 |
| | fast | 38 | 2.16 cm | 1.87 cm | **0.87×** [0.78, 1.00] | 0.83 |
| | all clips | 112 | 2.10 cm | 2.09 cm | **1.00×** [0.89, 1.16] | 0.33 |
| VGGT | slow | 37 | 0.87 cm | 0.82 cm | **0.95×** [0.89, 1.05] | 0.19 |
| | medium | 37 | 1.04 cm | 1.29 cm | **1.24×** [1.12, 1.50] | < 0.001 |
| | fast | 38 | 1.43 cm | 1.68 cm | **1.18×** [1.05, 1.26] | 0.00 |
| | all clips | 112 | 1.06 cm | 1.24 cm | **1.18×** [1.09, 1.26] | < 0.001 |
| π³ | slow | 37 | 0.56 cm | 0.56 cm | **1.00×** [0.85, 1.09] | 0.03 |
| | medium | 37 | 1.44 cm | 1.40 cm | **0.97×** [0.88, 1.21] | 0.03 |
| | fast | 38 | 0.76 cm | 1.16 cm | **1.51×** [0.83, 1.88] | 0.49 |
| | all clips | 112 | 0.78 cm | 0.88 cm | **1.12×** [0.90, 1.27] | 0.92 |

- Camera 2 is at most 1.0–1.2× worse when both cameras are normal. That explains only a small part of the 1.5–2.6× seen in the real test.
- Divided by the fairness ratio, the fast-motion penalty is still 1.17× (MapAnything), 1.25× (VGGT) and **2.33×** (π³).
- Caveat: the fairness recording films a calibration board, not the room, so it is a good check but not a perfect one.

### 3.7 Other checks

- **Timing of the answer key.** A rolling-shutter photo has no single capture time. We used the middle row's time (15 ms after the first row). Using the first row's time instead makes rolling shutter look

worse still: +1.6% (MapAnything), +2.6% (VGGT), +10% (π³). So our choice is the cautious one.

- **Turning vs walking.** In this dataset, fast-turning clips also tend to be fast-walking clips (correlation 0.40). Even “slow” clips walk at about 0.6 m/s and still have about 7.5 pixels of bend from turning. Extra statistics are in Appendix D.
- **Not a factor:** blur (same 2 ms exposure on both cameras), timing (photos paired exactly, 0.0 ms offset) and preparation (identical for both cameras).

### 3.8 The decision (pre-agreed rules)

| Result | Rule |
|---|---|
| **GO** | Fast motion ≥ 1.5× worse, AND it grows with speed, AND slow motion ≤ 1.2×. Must hold for MapAnything plus one other model. |
| **GRAY** | In between: run EXP1 and decide there. |
| **STOP** | Under 1.2× in fast motion for all models: drop as a main paper. |

| Model | Fast motion | Grows with speed? | Slow motion ≤ 1.2×? | Result |
|---|---|---|---|---|
| MapAnything | 1.16× | barely | yes (1.14×) | STOP |
| VGGT | 1.47× | no (turning error: yes) | no (1.73×) | GRAY |
| π³ | **2.61×** | **yes, clearly** | no (1.70×) | GRAY |

**EXP0 overall: GRAY.** The effect is large for VGGT and π³, but even slow clips were worse, so EXP0 alone could not prove that the bending is the cause. EXP1 was designed to settle that.

### 3.9 What EXP0 proved, and what it could not

- **Proved:** rolling shutter makes VGGT and π³ clearly worse. For π³ it gets worse with speed. It is not just “camera 2 is worse”.
- **Could not prove:** that the bending is the real cause (the slow-clip puzzle), and how much accuracy we could win back by fixing it.

# 4. EXP1: can it be fixed if we know the motion?

> If we straighten every rolling-shutter photo using the **true** camera motion, how much of the extra error goes away? **Pre-agreed rule:** GO if at least 50% goes away in fast motion. STOP if under 25%.

## 4.1 How straightening works

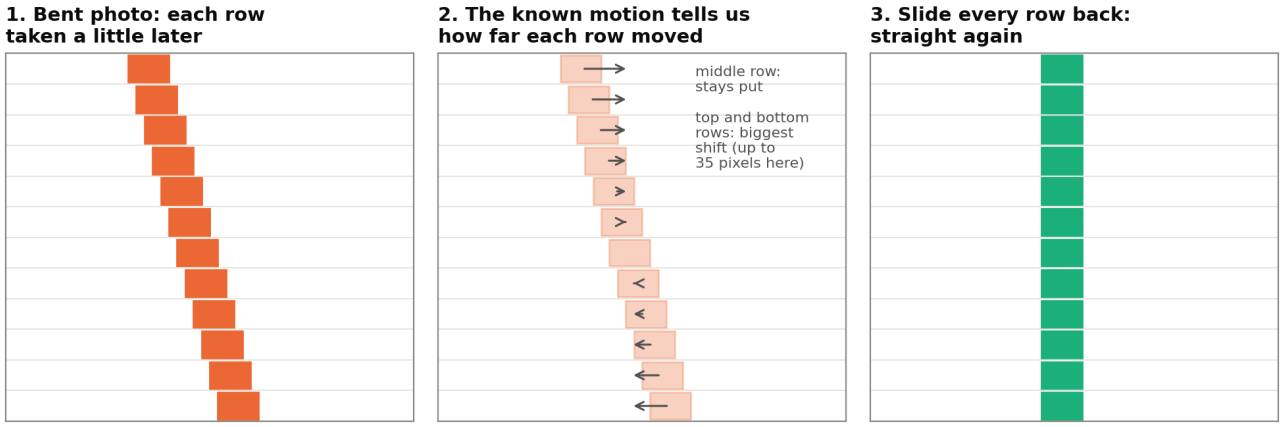



*Figure 7. Straightening, step by step. Every row is moved to where it would have been at the moment the middle row was captured.*

1. **E1a, the planned test (turning only):** the gyroscope says how fast the camera was turning during the 30 ms. From that we compute, for every row, how far it moved, and slide it back. No depth needed.
2. **E1b, the extra test we recommended after EXP0 (turning + walking):** walking also bends the photo, but how much depends on how far away each object is. Near things shift more. So E1b also needs distances. We used π³'s depth estimate, scaled to metres with the true path, plus the true walking speed.
3. The straightening is done on the raw lens image, before the usual lens correction. That keeps each row's timing exact. The rest of the preparation is identical to EXP0.
4. All three models are re-run on the straightened photos: same 1,124 clips, same scoring, same answer key.

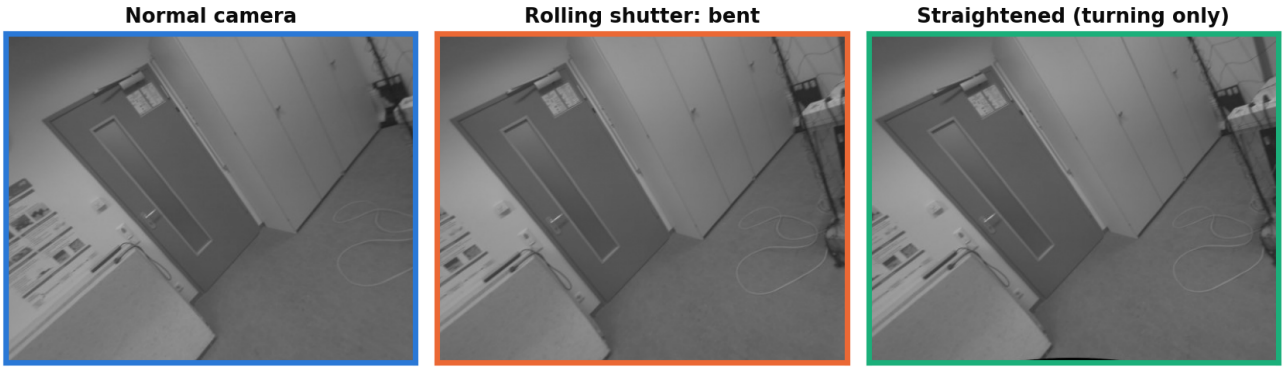



*Figure 8. The fastest frame (about 180°/s): normal camera, bent rolling-shutter photo, and straightened. Look at the door frame and the cupboard edge. This check picture was made before we switched how edges are filled, so a thin black sliver shows at the bottom right. In the real run, pixels from beyond the sensor edge repeat the edge pixel instead.*

## 4.2 Safety checks (all passed before the run)

| Check | Result |
|---|---|
| With zero motion, is the photo unchanged? | Yes: identical to EXP0's photo (0.0 pixel difference). |
| Are the gyroscope's directions right? | The predicted rotation over one photo matches the true rotation within 0.13° (out of 4.45°). A sign mistake would show up as 8.96°. |
| How big are the shifts? | Up to 35 pixels, at 3.1 rad/s (about 180°/s). |
| Does it look right? | Yes: the straightened photo lines up with the normal camera's (Figure 8). |

## 4.3 Main result: share of the extra error removed

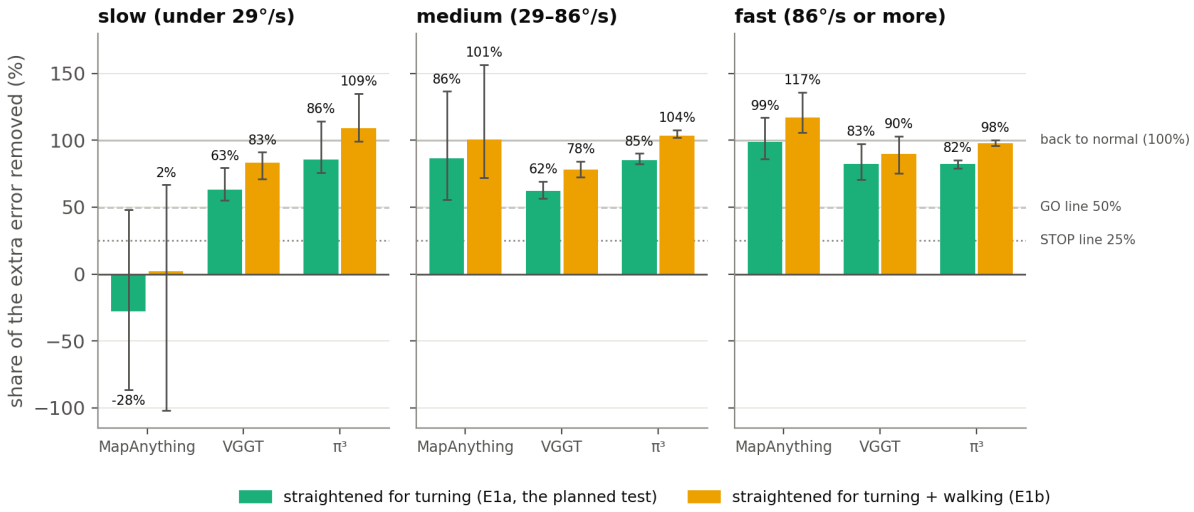



*Figure 9. Share of the extra rolling-shutter error removed, by speed group. Thin lines show the 95% range. MapAnything's slow-group bars are noise: its gap there is only 0.5 cm.*

## Table 4.1. Path error (cm) and share of the extra error removed

| Model | Speed | Normal (cm) | Bent (cm) | Fixed for turning (cm) | Share removed [95%] | Fixed for turning + walking (cm) | Share removed [95%] |
|---|---|---|---|---|---|---|---|
| MapAnything | slow | 3.42 | 3.89 | 4.02 | **-28%** [-86, 48] | 3.88 | **2%** [-102, 67] |
| | medium | 6.60 | 7.01 | 6.66 | **86%** [56, 137] | 6.60 | **101%** [72, 157] |
| | fast | 9.33 | 10.85 | 9.35 | **99%** [86, 117] | 9.07 | **117%** [106, 136] |
| | all clips | 7.48 | 8.17 | 7.38 | **115%** [95, 135] | 7.33 | **123%** [105, 143] |
| VGGT | slow | 1.36 | 2.35 | 1.73 | **63%** [55, 79] | 1.52 | **83%** [71, 91] |
| | medium | 2.29 | 3.86 | 2.88 | **62%** [56, 69] | 2.63 | **78%** [73, 84] |
| | fast | 6.60 | 9.72 | 7.14 | **83%** [71, 97] | 6.91 | **90%** [75, 103] |
| | all clips | 3.28 | 5.73 | 3.93 | **74%** [69, 79] | 3.75 | **81%** [77, 86] |
| π³ | slow | 0.77 | 1.32 | 0.85 | **86%** [76, 114] | 0.72 | **109%** [99, 135] |
| | medium | 1.03 | 2.26 | 1.21 | **85%** [82, 90] | 0.99 | **104%** [102, 108] |
| | fast | 1.95 | 5.10 | 2.51 | **82%** [79, 85] | 2.03 | **98%** [96, 100] |
| | all clips | 1.31 | 3.29 | 1.59 | **86%** [83, 89] | 1.27 | **102%** [101, 104] |

Values above 100% mean the straightened photos scored slightly better than the normal camera. These differences are within the camera-to-camera spread measured by the fairness check, so read them as “back to normal level”.

## 4.4 The same thing in centimetres

Orange (bent) is always highest; once straightened (aqua, yellow) the error drops back to blue (normal). Each panel has its own scale.

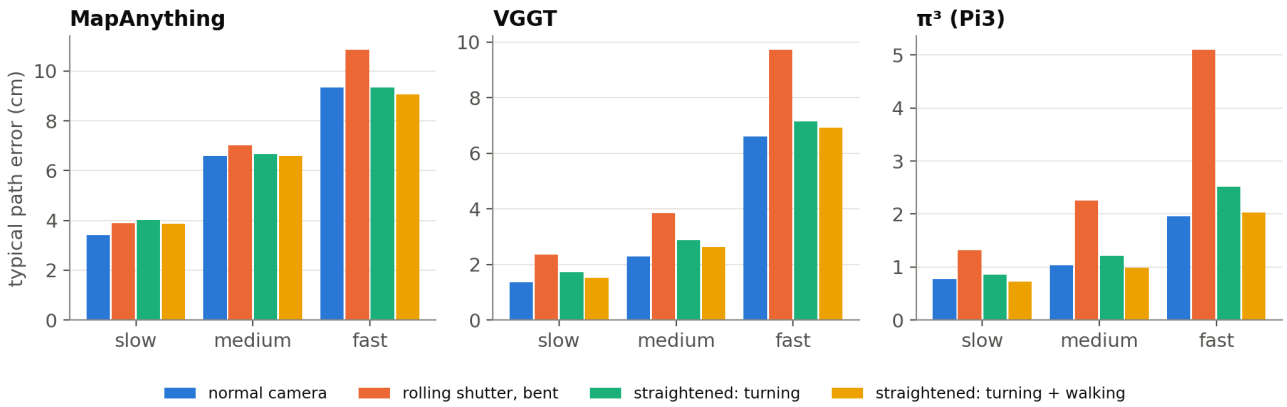



*Figure 10. Typical path error for each version of the photos.*

## 4.5 Does it work at every speed?

The fix works at every speed, not only for fast motion. Dotted = 25% (STOP), dashed = 50% (GO), top line = 100% (back to normal).

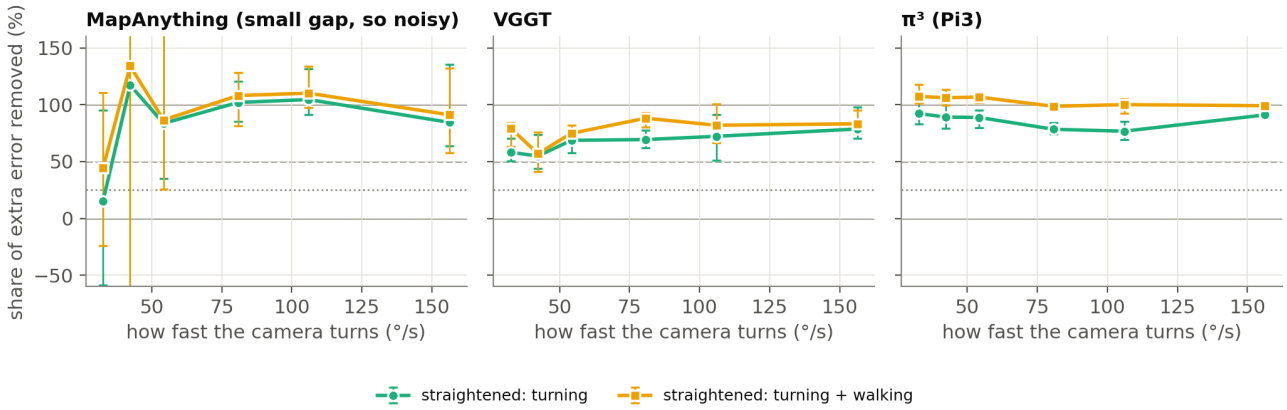



*Figure 11. Share removed against turning speed (six equal-size groups of clips per model). The share is flat: the fix works as well for slow clips as for fast ones.*

## 4.6 The fastest clip again

The fastest 2-second clip (about 200°/s): path error on each version of the same 16 photos

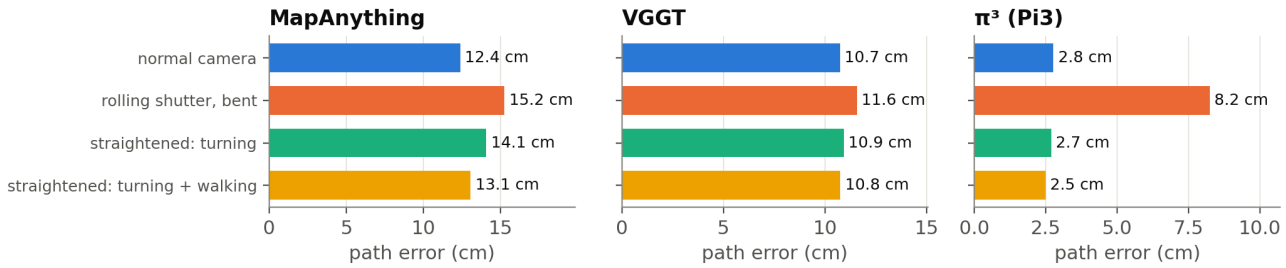



*Figure 12. The same fastest clip as Figure 5. π³: 2.8 cm (normal), 8.2 cm (bent), 2.7 cm (straightened for turning), 2.5 cm (turning + walking).*

| Model | Normal | Bent | Straightened: turning | Straightened: turning + walking |
|---|---|---|---|---|
| MapAnything | 12.4 cm | 15.2 cm | 14.1 cm | 13.1 cm |
| VGGT | 10.7 cm | 11.6 cm | 10.9 cm | 10.8 cm |
| π³ | 2.8 cm | 8.2 cm | 2.7 cm | 2.5 cm |

## 4.7 How consistent is the improvement? (Table 4.2)

| Model | Speed | Clips | Improved: turning | Improved: turning + walking | p: turning | p: turning + walking |
|---|---|---|---|---|---|---|
| MapAnything | slow | 56 | 39% | 55% | 0.16 | 0.97 |
| | medium | 648 | 67% | 65% | < 0.001 | < 0.001 |
| | fast | 420 | 80% | 77% | < 0.001 | < 0.001 |
| | all clips | 1124 | 71% | 69% | < 0.001 | < 0.001 |
| VGGT | slow | 56 | 96% | 93% | < 0.001 | < 0.001 |
| | medium | 648 | 95% | 93% | < 0.001 | < 0.001 |
| | fast | 420 | 87% | 86% | < 0.001 | < 0.001 |
| | all clips | 1124 | 92% | 90% | < 0.001 | < 0.001 |
| π³ | slow | 56 | 100% | 98% | < 0.001 | < 0.001 |
| | medium | 648 | 98% | 99% | < 0.001 | < 0.001 |
| | fast | 420 | 86% | 85% | < 0.001 | < 0.001 |
| | all clips | 1124 | 94% | 94% | < 0.001 | < 0.001 |

“Improved” = share of clips where the straightened photos gave a smaller path error than the bent ones. p compares straightened against bent, clip by clip.

## 4.8 The other two scores (Tables 4.3 and 4.4)

| Model | Speed | Normal (°) | Bent (°) | Fixed for turning (°) | Share removed [95%] | Fixed for turning + walking (°) | Share removed [95%] |
|---|---|---|---|---|---|---|---|
| MapAnything | slow | 0.37 | 0.37 | 0.35 | — (no gap) | 0.33 | — (no gap) |
| | medium | 0.56 | 0.61 | 0.54 | **127%** [104, 189] | 0.54 | **133%** [113, 189] |
| | fast | 1.22 | 1.54 | 1.17 | **117%** [104, 135] | 1.16 | **122%** [107, 137] |
| | all clips | 0.74 | 0.88 | 0.72 | **115%** [104, 135] | 0.71 | **127%** [115, 144] |
| VGGT | slow | 0.22 | 0.32 | 0.23 | **84%** [74, 98] | 0.23 | **81%** [71, 100] |
| | medium | 0.30 | 0.46 | 0.32 | **87%** [82, 91] | 0.31 | **90%** [85, 93] |
| | fast | 0.67 | 1.14 | 0.72 | **89%** [84, 94] | 0.71 | **93%** [89, 98] |
| | all clips | 0.40 | 0.62 | 0.42 | **89%** [84, 92] | 0.42 | **92%** [89, 96] |
| π³ | slow | 0.18 | 0.24 | 0.18 | **104%** [80, 148] | 0.18 | **99%** [78, 146] |
| | medium | 0.22 | 0.36 | 0.23 | **97%** [92, 102] | 0.22 | **100%** [96, 105] |
| | fast | 0.42 | 0.76 | 0.43 | **99%** [94, 102] | 0.42 | **102%** [99, 105] |
| | all clips | 0.28 | 0.46 | 0.29 | **96%** [93, 101] | 0.28 | **102%** [98, 106] |

*Table 4.3. Turning error (degrees). “No gap” = rolling shutter was within 5% of normal, so there was nothing to remove.*

| Model | Speed | Normal (cm) | Bent (cm) | Fixed for turning (cm) | Share removed [95%] | Fixed for turning + walking (cm) | Share removed [95%] |
|---|---|---|---|---|---|---|---|
| MapAnything | slow | 2.40 | 3.28 | 3.12 | **18%** [-27, 34] | 3.00 | **31%** [-13, 55] |
| | medium | 4.99 | 5.51 | 5.09 | **81%** [56, 103] | 5.05 | **88%** [64, 118] |
| | fast | 9.09 | 10.62 | 9.20 | **93%** [79, 107] | 9.20 | **93%** [83, 112] |
| | all clips | 5.94 | 6.83 | 6.24 | **66%** [52, 80] | 6.18 | **73%** [59, 92] |
| VGGT | slow | 0.91 | 1.80 | 1.11 | **78%** [74, 85] | 0.99 | **91%** [85, 97] |
| | medium | 1.48 | 2.73 | 1.83 | **72%** [68, 78] | 1.73 | **80%** [75, 86] |
| | fast | 4.64 | 7.33 | 5.31 | **75%** [70, 81] | 4.98 | **87%** [82, 92] |
| | all clips | 2.28 | 4.06 | 2.60 | **82%** [78, 86] | 2.51 | **87%** [83, 91] |
| π³ | slow | 0.69 | 1.22 | 0.69 | **99%** [89, 112] | 0.66 | **105%** [96, 118] |
| | medium | 0.84 | 1.84 | 0.88 | **95%** [92, 98] | 0.81 | **103%** [100, 106] |
| | fast | 1.79 | 3.84 | 1.96 | **92%** [89, 94] | 1.80 | **100%** [97, 102] |
| | all clips | 1.13 | 2.37 | 1.27 | **89%** [87, 92] | 1.11 | **102%** [100, 105] |

*Table 4.4. Step error (cm).*

### 4.9 The decision

| Model | Fast clips: turning only (planned test) | Fast clips: turning + walking | Result |
|---|---|---|---|
| MapAnything | **99%** [86, 117] | 117% [106, 136] | **GO** |
| VGGT | **83%** [71, 97] | 90% [75, 103] | **GO** |
| π³ | **82%** [79, 85] | 98% [96, 100] | **GO** |

> **EXP1 overall: GO** for all three models, in both versions. Even the bottom of every 95% range is well above the 50% GO line.

### 4.10 What EXP1 proved, and what it did not

- **Proved: the bending is the whole cause.** Once the photos are straightened, accuracy goes back to the normal camera's level, at every speed.
- **Solved EXP0's slow-clip puzzle.** The slow-clip penalty was also rolling shutter, mostly from walking: slow-clip share removed for π³ 86% → 109%, VGGT 63% → 83% when walking is added.
- **Walking matters, not just turning.** Adding walking gains 15–20 points (π³ fast 82% → 98%). Our method should predict both.
- **Did not prove that a real system can do this.** E1a used a real gyroscope. E1b used the true path and true scale.
- **Did not prove it generalizes.** One room, one camera rig, one readout time (30 ms).

## 5. Where we stand

| We now know (measured) | We don't know yet |
|---|---|
| Rolling shutter hurts VGGT and π³ a lot (1.5–2.6× in fast motion) and MapAnything a little (1.16×). | Can a model estimate the motion from images well enough to fix itself? **(EXP2)** |
| The damage grows with speed (π³: 1.6× → 3.9×). | Does a trained “motion head” beat simple fixes? **(EXP3)** |
| It is caused by rolling shutter, not camera differences (those are at most 1.2×). | Does it hold on phones, other readout times and outdoor scenes? |
| Knowing the motion removes 82–100% of the damage, at every speed. | Has someone already published this exact idea? **(novelty check)** |
| Both turning and walking matter. | |

## 6. The road ahead, and why

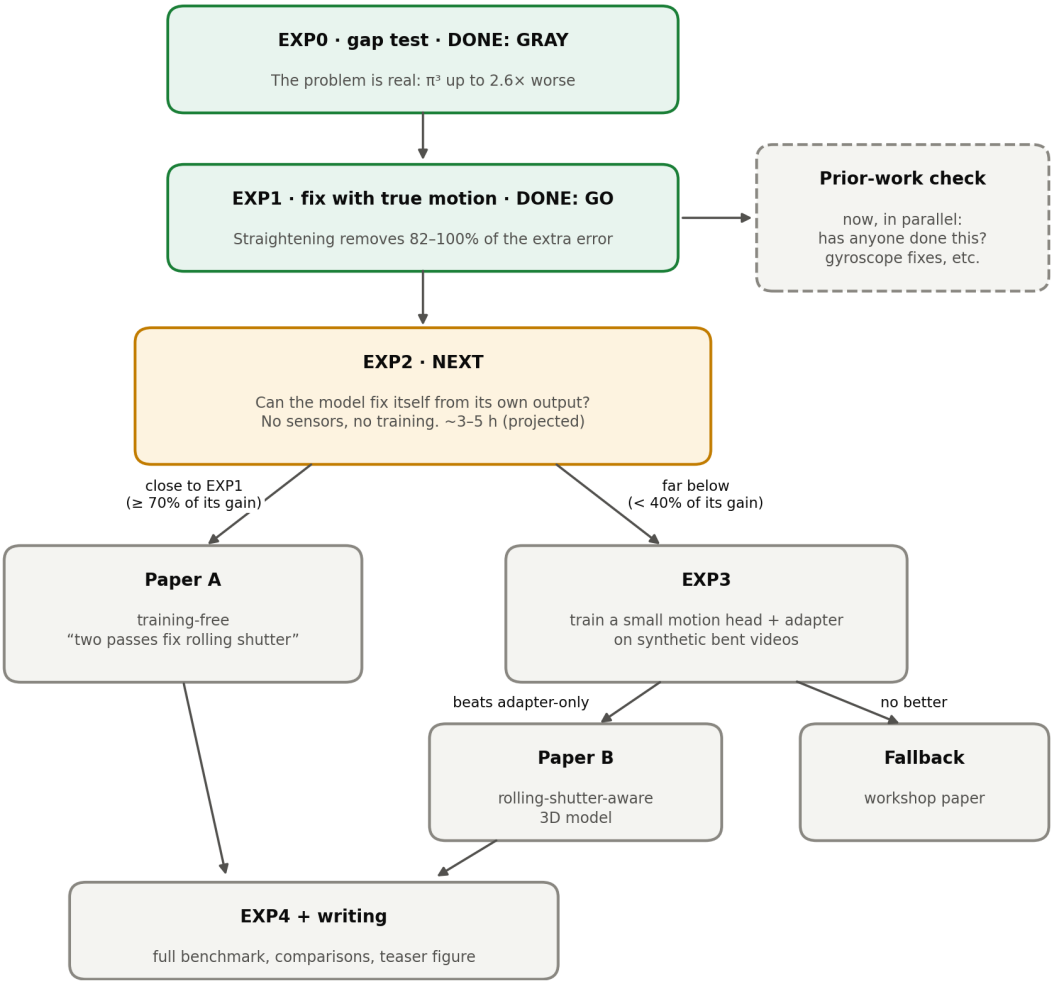



*Figure 13. The plan from here. Green = done, amber = next, grey = later. The decision lines (70% / 40%) are a proposal for the team to confirm before EXP2 runs.*

### Deadline

CVPR 2027: registration around **10 November** and submission around **16 November 2026** (from the project brief; please verify on the CVPR site). As of 29 October that is about **12 days to registration and 18 days to submission**. It is tight but workable, since each experiment so far took about 2 hours of computer time.

### Step 1: prior-work check (now, alongside EXP2)

- **What:** search recent arXiv (last 60 days) and the main venues for rolling shutter combined with feed-forward 3D models. Also search for gyroscope-based rolling-shutter correction, SMARSC (CVPR 2019) and rolling-shutter unrolling networks.
- **Why:** similar adapters for fisheye lenses were published recently (RayTun3R, Fisheye3R), so rolling shutter is the obvious next target and scoop risk is real. Reviewers will also ask “why not just use the gyroscope?”, so we must know exactly what already exists.

### Step 2: EXP2, the cheap training-free fix (next; projected 3–5 hours including coding)

1. Run the model once on the bent clip.
2. From its own answer, estimate how the camera was turning and moving: compare neighbouring photos' positions and divide by the time between them.
3. Straighten the photos with that estimate and run the model again.
4. Compare against EXP1 (the best case) and against the bent photos.

- **Why it is next:** it decides which paper we write, it answers the “no gyroscope” question directly, and it costs very little because EXP1's code is reused. We need it as a comparison in the final paper anyway.
- **A helpful detail:** the bend from walking depends on speed ÷ distance. The model gives both in the same unknown units, so the units cancel and EXP2 can correct walking too, without knowing real-world scale.
- **Expected difficulty:** neighbouring photos in a clip are 150 ms apart, while a photo takes 30 ms, and the first-pass answer is itself bent. So the estimate will be rougher than the gyroscope's. EXP2 measures how much that matters.
- **Decision lines (proposal, agree before running):** “close to EXP1” = EXP2 achieves at least 70% of EXP1's gain in fast clips. “Far below” = under 40%. In between: report EXP2 as a strong baseline and also build EXP3.

### Step 3: depends on EXP2

- **If EXP2 is close to EXP1, Paper A:** a training-free paper, “two passes fix rolling shutter in any 3D model”. Cheaper and lower-risk.
- **If EXP2 falls short, EXP3:** build a generator of synthetic bent videos (from ScanNet++ / TartanAir, which have depth and camera paths). Train a small motion head plus a light adapter (LoRA) that predicts turning and walking per photo. GO only if it clearly beats the same adapter trained *without* the motion head, and does not hurt normal photos. Projected 4–6 h for the first run, plus building the generator.

### Step 4: full benchmark and writing

Compare: plain model; “straighten the photo with an existing unrolling network, then reconstruct”; gyroscope straightening (E1a, already done); EXP2; adapter only; ours. Measure path error on TUM, 3D shape quality on synthetic rolling-shutter data (as far as we know, TUM has camera paths only, no 3D shape ground truth) and no loss on normal photos. Teaser figure: a fast pan where the plain model bends a wall and ours keeps it straight.

## Recommended changes to the original plan

| Change | Why (evidence) |
|---|---|
| **Main model: π³, then VGGT.** MapAnything becomes secondary (the plan had it first). | π³ has the largest gap (2.6×) and the tightest results ([79, 85]%). MapAnything's gap is so small that gains are hard to show. |
| **Predict both turning and walking**, kept small by regularization (the plan said “focus on turning”). | E1b: walking adds 15–20 points and explains the slow-clip penalty. |
| **Add gyroscope straightening as a baseline.** | E1a shows it works very well. Reviewers will ask, and we already have the numbers. |
| **Speed up the pipeline before EXP2.** | Most run time was spent on the CPU (straightening, loading photos), not the GPU. *Projected (untested):* caching and GPU straightening could cut run time by half or more. |

## Risks

| Risk | How likely | What we do |
|---|---|---|
| Someone publishes the same idea first | Medium (fisheye versions are out) | Re-search arXiv weekly, move fast, keep the workshop as a fallback. |
| “Just use a gyroscope” | High, as a reviewer question | Images-only is our angle. Include the gyroscope baseline. Stress videos with no sensor data. |
| The model's own motion estimate is too rough | Unknown | EXP2 tests exactly this. If too rough, a learned motion head is the paper (EXP3). |
| Only one dataset so far | Certain right now | Synthetic rolling-shutter data with different readout times. A second real dataset if one exists. |
| Walking speed is hard to estimate | Medium | Regularize toward zero. Report turning-only and turning + walking separately. |
| Deadline | Tight (~18 days) | Each step has a GO/STOP rule. The workshop track is the fallback. |

# Appendix A. Glossary

| Term | Meaning |
|---|---|
| Rolling shutter (RS) | Camera reads the photo row by row (here over 30.18 ms). |
| Global shutter (GS), “normal” | Camera captures every row at the same instant. |
| Clip (window) | 16 photos from 2.25 s of a recording. |
| Path error (ATE) | Average distance between the model's camera positions and the true ones, after lining up and scaling (cm). |
| Turning error (RPE-rot) | Error in how much the camera turned between neighbouring photos (degrees). |
| Step error (RPE-trans) | Error in how far the camera moved between neighbouring photos (cm). |
| Straightening (unrolling) | Moving each row of a rolling-shutter photo to where it would be if the whole photo had been taken at once. |
| Oracle | Using information a real system would not have (here: gyroscope and true path), to measure the best case. |
| Share of extra error removed | (bent − straightened) ÷ (bent − normal), using medians. |
| Fairness-adjusted share | The same, but measured against the normal camera's error × the fairness ratio (how much worse camera 2 is when both cameras are normal). |
| 95% range | Bootstrap confidence interval: clips resampled 1,000 times. |
| p (Wilcoxon test) | Clip-by-clip test of whether one version is consistently better than the other. |
| rad/s | Radians per second. 1 rad/s ≈ 57°/s. |
| LoRA / adapter | A small add-on trained on top of a frozen model, so the big model is not retrained. |

# Appendix B. Run facts

| | EXP0 | EXP1 |
|---|---|---|
| Total time | about 1 h 50 min (setup 25 min, photo preparation 5 min, model runs about 70 min, analysis 5 min) | 2 h 02 min (17:14 to 19:15 UTC) |
| Photo work | 1,124 clips × 2 cameras, plus 112 fairness clips | 9,248 photos straightened per version (turning: 14 min; turning + walking: 17 min, on the CPU). π³ depth for E1b: 65 min. |
| Model run time | MapAnything 58 min, VGGT about 70 min, π³ about 78 min (2,248 clips each, side by side) | Turning: MapAnything 39.0, VGGT 46.1, π³ 48.2 min. Turning + walking: 31.9, 37.6, 39.0 min (1,124 clips each, side by side). |
| GPU memory (peak per model) | MapAnything 13.5 GB, VGGT 7.5 GB, π³ 6.1 GB | same as EXP0 |
| Shared GPU (80 GB) | Peak use 38.7 GB. Free memory never below about 42 GB. | Free memory never below 26.9 GB (rule: 12 GB). Our processes were set to low CPU priority so the other team's job kept first pick. |
| Settings | bf16 precision, one clip at a time per model, photos only | same as EXP0 |

# Appendix C. Changes from the plan

## EXP0

- π³'s code moved to a new address (github.com/yyfz/Pi3). The same official code and weights were used.
- The three models ran side by side to save time. MapAnything's optional output masks were turned off for speed; this does not affect camera positions.
- Extras not in the plan: the fairness recording, finer speed steps, the timing-sensitivity check, and the turning/walking statistics.

## EXP1

- Straightening is done on the raw lens image, not with the plan's simple formula on a corrected image. It is the same idea, but exact for this fisheye-like lens.
- Turning comes from the gyroscope (averaged over the 30 ms of each photo). The true path was used only to check it.
- Added E1b (turning + walking). It uses the true walking speed and π³'s depth scaled with the true path. π³'s depth was used for all three models, which may slightly favour π³ in E1b.
- Pixels shifted in from beyond the sensor edge repeat the edge pixel instead of being black, so a black border cannot give the models a hint. At fast motion this leaves smeared strips up to about 35 pixels wide. That can only hurt the straightened result, so the shares removed are, if anything, conservative.

# Appendix D. Extra statistics

## D.1 Does the penalty grow with turning speed? (EXP0, all 1,124 clips)

| Model | Score | Trend with turning speed (Spearman ρ) | p |
|---|---|---|---|
| MapAnything | path error | +0.23 | < 0.001 |
| MapAnything | turning error | +0.28 | < 0.001 |
| MapAnything | step error | -0.04 | 0.18 |
| VGGT | path error | -0.18 | < 0.001 |
| VGGT | turning error | +0.11 | < 0.001 |
| VGGT | step error | -0.37 | < 0.001 |
| π³ | path error | +0.25 | < 0.001 |
| π³ | turning error | +0.30 | < 0.001 |
| π³ | step error | -0.00 | 0.91 |

Positive ρ = the rolling-shutter penalty grows with turning speed. π³ and MapAnything: yes for path and turning error. VGGT: yes for turning error only.

## D.2 Turning vs walking: which drives the penalty? (EXP0)

| Model | Score | Effect of turning speed (per rad/s) | Effect of walking speed (per m/s) | Times worse with no motion (fitted) |
|---|---|---|---|---|
| MapAnything | path error | +0.050 ± 0.009 | +0.005 ± 0.019 | 1.03× |
| MapAnything | turning error | +0.052 ± 0.009 | +0.059 ± 0.019 | 1.03× |
| VGGT | path error | -0.096 ± 0.013 | +0.067 ± 0.028 | 1.73× |
| VGGT | turning error | +0.098 ± 0.011 | -0.200 ± 0.023 | 1.71× |
| π³ | path error | +0.206 ± 0.021 | -0.234 ± 0.044 | 2.08× |
| π³ | turning error | +0.172 ± 0.013 | -0.145 ± 0.028 | 1.55× |

A straight-line fit of log(times worse) on turning speed and walking speed together. Turning speed and walking speed are correlated (0.40), so read these as rough. For π³, turning clearly drives the penalty.

## D.3 Direction-of-travel accuracy (EXP0): share of photo pairs with direction off by less than 5°

| Model | Speed | Normal camera | Rolling shutter | Rolling ÷ normal |
|---|---|---|---|---|
| MapAnything | slow | 34% | 31% | 0.91× |
| | medium | 29% | 28% | 0.96× |
| | fast | 28% | 19% | 0.68× |
| VGGT | slow | 85% | 62% | 0.73× |
| | medium | 78% | 45% | 0.57× |
| | fast | 36% | 18% | 0.49× |
| π³ | slow | 91% | 80% | 0.88× |
| | medium | 97% | 77% | 0.79× |
| | fast | 89% | 52% | 0.59× |

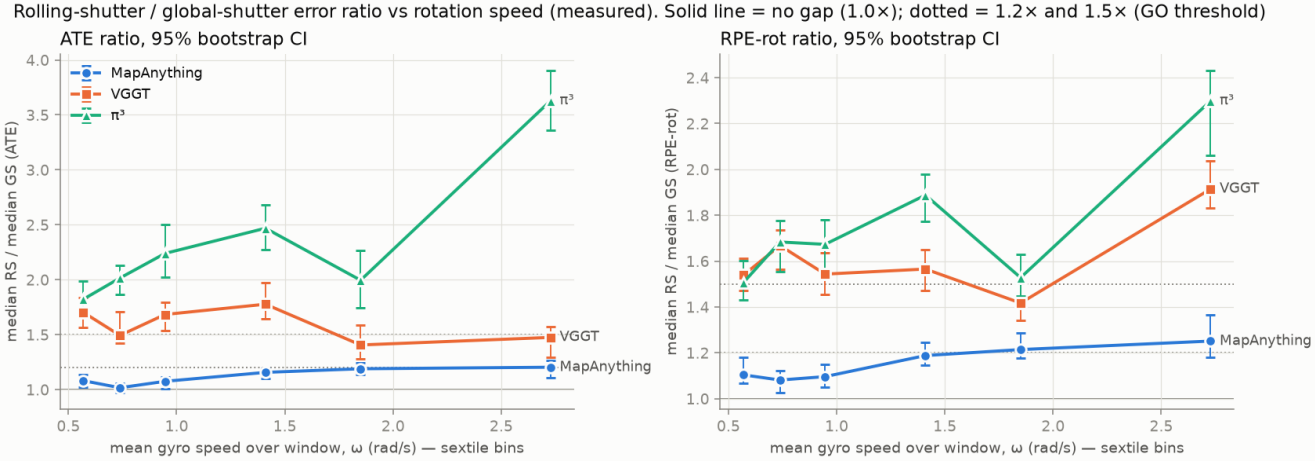
Higher is better. For π³ in fast motion it drops from 89% to 52% on rolling-shutter photos. Turning accuracy (the same test for rotation) had a median of 97–100% on both cameras for every model and speed, so it is not shown.

## D.4 EXP1 share removed, plain vs fairness-adjusted (path error)

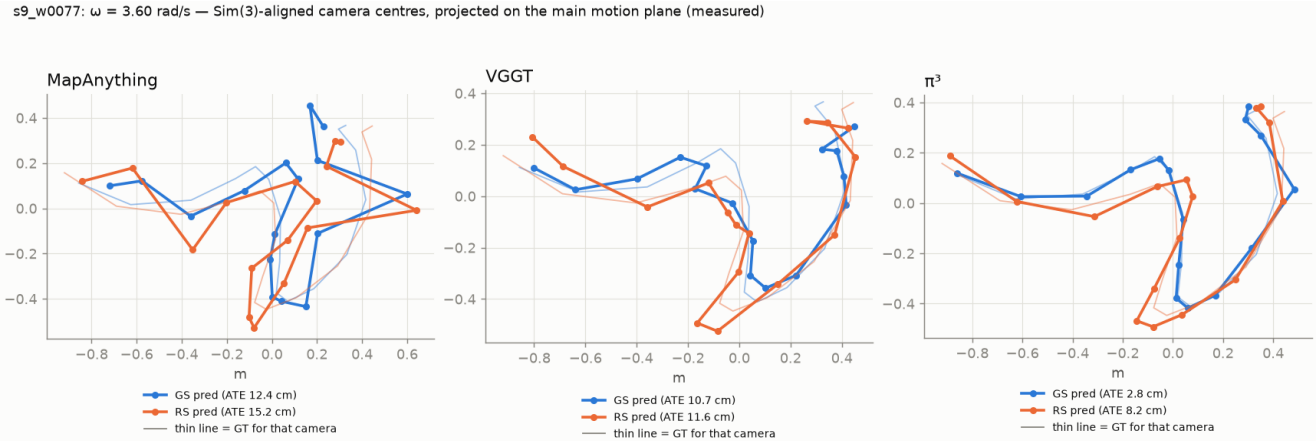
| Model | Speed | Turning: plain | Turning: fairness-adjusted | Turning + walking: plain | Turning + walking: fairness-adjusted |
|---|---|---|---|---|---|
| MapAnything | slow | -28% | -28% | 2% | 2% |
| | medium | 86% | 84% | 101% | 98% |
| | fast | 99% | 98% | 117% | 116% |
| | all clips | 115% | 113% | 123% | 120% |
| VGGT | slow | 63% | 83% | 83% | 109% |
| | medium | 62% | 84% | 78% | 105% |
| | fast | 83% | 131% | 90% | 143% |
| | all clips | 74% | 96% | 81% | 106% |
| $\pi^3$ | slow | 86% | 104% | 109% | 132% |
| | medium | 85% | 95% | 104% | 115% |
| | fast | 82% | 89% | 98% | 105% |
| | all clips | 86% | 94% | 102% | 111% |

Fairness-adjusted: the target is the normal camera's error × the fairness ratio (1.00 / 1.18 / 1.12), because camera 2 is slightly worse even without rolling shutter. MapAnything's slow numbers are noise (tiny gap).

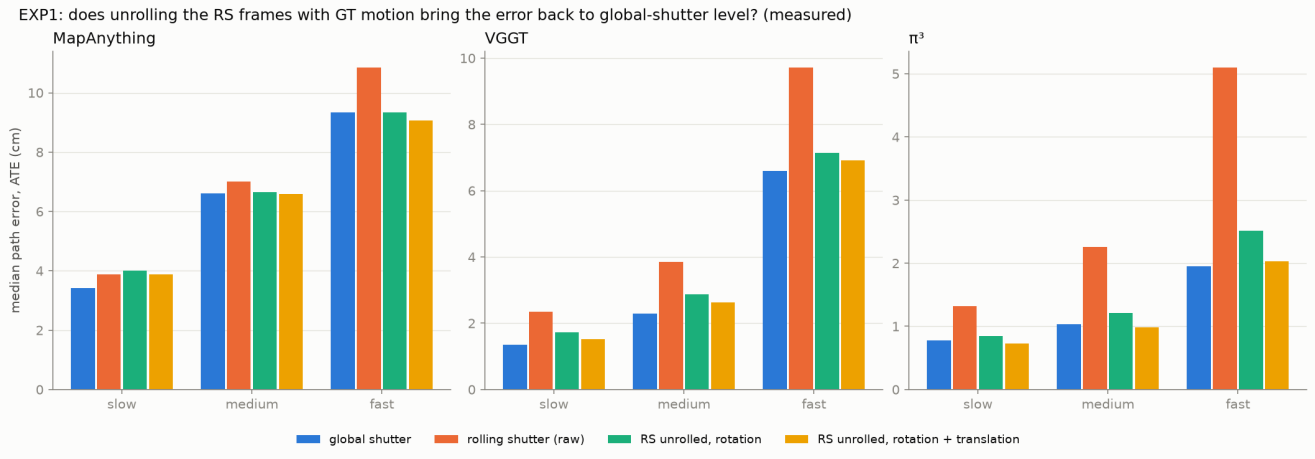
# Appendix E. Technical charts (as produced by the analysis code)



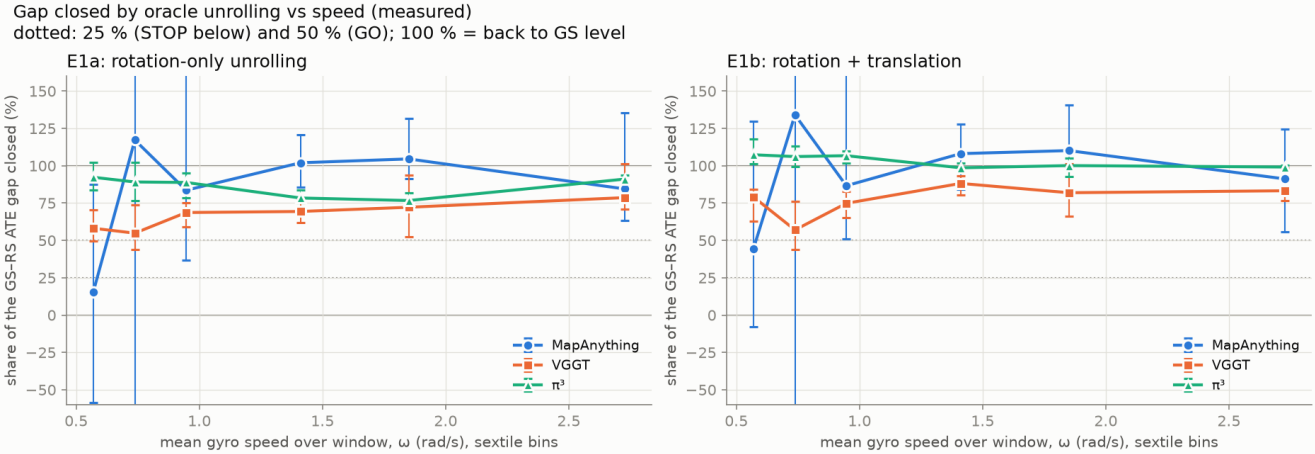

*E.1 EXP0: times worse against turning speed, six equal-size speed groups, with 95% ranges.*




*E.2 EXP0: top-down camera paths for the fastest clip, all three models.*




*E.3 EXP1: median path error per speed group for normal, bent, straightened (turning) and straightened (turning + walking).*




*E.4 EXP1: share removed against turning speed, both versions, all models.*

# Appendix F. Files

| Where | What |
|---|---|
| `EXPERIMENTS_SO_FAR.md` | Short written summary of both experiments and the road ahead. |
| `results/exp0/` | summary.md, DATA_NOTES.md, all tables (CSV) and charts. |
| `results/exp1/` | summary.md, e1_summary.csv, per_window_e1.csv (every clip), charts. |
| `src/` | All code: preparation, model runs, straightening, analysis. |
| `report/` | This report, the earlier EXP0 report, and the scripts that build them. |
| Only on the GPU box | EXP1's raw model outputs, straightened photos and depth maps. They can be rebuilt in about 2 hours. Every score is already saved locally. |